Experiment 4

SQL Join Operations
Student and Dept tables | Oracle Database 21c XE

1. Source Code

```
-- 4-1. Create the dept table having dno and dname
CREATE TABLE dept (
  dno   NUMBER,
  dname VARCHAR2(20)
);

DESC dept;

-- 4-2. Apply the primary key to dno and NOT NULL to dname
ALTER TABLE dept ADD CONSTRAINT pk_dept PRIMARY KEY (dno);
ALTER TABLE dept MODIFY dname NOT NULL;

DESC dept;

-- 4-3. Create the student table having sid, sname and did
CREATE TABLE student (
  sid   NUMBER,
  sname VARCHAR2(30),
  did   NUMBER
);

DESC student;

-- 4-4. Apply primary key to sid, NOT NULL to sname and foreign key to did
ALTER TABLE student ADD CONSTRAINT pk_student PRIMARY KEY (sid);
ALTER TABLE student MODIFY sname NOT NULL;
ALTER TABLE student ADD CONSTRAINT fk_student_dept
  FOREIGN KEY (did) REFERENCES dept(dno);

DESC student;

-- 4-5. Insert the department details
INSERT INTO dept VALUES (10,'CSE');
INSERT INTO dept VALUES (20,'ME');
INSERT INTO dept VALUES (30,'CE');
INSERT INTO dept VALUES (40,'EEE');
INSERT INTO dept VALUES (50,'ECE');
INSERT INTO dept VALUES (60,'CSM');
INSERT INTO dept VALUES (70,'CSD');
COMMIT;

SELECT * FROM dept;

-- 4-6. Insert 10 rows into the student table
INSERT ALL
  INTO student VALUES (1,'Ravi',10)
  INTO student VALUES (2,'Sita',10)
  INTO student VALUES (3,'Arun',20)
  INTO student VALUES (4,'Latha',30)
  INTO student VALUES (5,'Kiran',40)
  INTO student VALUES (6,'Meena',50)
  INTO student VALUES (7,'Varun',20)
  INTO student VALUES (8,'Divya',30)
  INTO student VALUES (9,'Hari',10)
  INTO student VALUES (10,'Anu',NULL)
SELECT * FROM dual;

COMMIT;
SELECT * FROM student;

-- 4-7. NATURAL JOIN between student and dept
SELECT * FROM student
NATURAL JOIN (SELECT dno AS did, dname FROM dept);

-- 4-8. EQUI JOIN between student and dept
SELECT s.sid, s.sname, s.did, d.dno, d.dname
FROM student s
JOIN dept d ON s.did = d.dno;
```

```sql
-- 4-9. CONDITIONAL JOIN between student and dept
SELECT s.sid, s.sname, s.did, d.dno, d.dname
FROM student s
JOIN dept d ON s.did = d.dno AND d.dname <> 'CSE';

-- 4-10. LEFT OUTER NATURAL JOIN
SELECT * FROM student
NATURAL LEFT OUTER JOIN (SELECT dno AS did, dname FROM dept);

-- 4-11. RIGHT OUTER NATURAL JOIN
SELECT * FROM student
NATURAL RIGHT OUTER JOIN (SELECT dno AS did, dname FROM dept);

-- 4-12. FULL OUTER NATURAL JOIN
SELECT * FROM student
NATURAL FULL OUTER JOIN (SELECT dno AS did, dname FROM dept);

-- 4-13. LEFT OUTER EQUI JOIN
SELECT s.sid, s.sname, s.did, d.dno, d.dname
FROM student s
LEFT OUTER JOIN dept d ON s.did = d.dno;

-- 4-14. RIGHT OUTER EQUI JOIN
SELECT s.sid, s.sname, s.did, d.dno, d.dname
FROM student s
RIGHT OUTER JOIN dept d ON s.did = d.dno;

-- 4-15. FULL OUTER EQUI JOIN
SELECT s.sid, s.sname, s.did, d.dno, d.dname
FROM student s
FULL OUTER JOIN dept d ON s.did = d.dno;

-- 4-16. LEFT OUTER CONDITIONAL JOIN
SELECT s.sid, s.sname, s.did, d.dno, d.dname
FROM student s
LEFT OUTER JOIN dept d ON s.did = d.dno AND d.dname <> 'CSE';

-- 4-17. RIGHT OUTER CONDITIONAL JOIN
SELECT s.sid, s.sname, s.did, d.dno, d.dname
FROM student s
RIGHT OUTER JOIN dept d ON s.did = d.dno AND d.dname <> 'CSE';

-- 4-18. FULL OUTER CONDITIONAL JOIN
SELECT s.sid, s.sname, s.did, d.dno, d.dname
FROM student s
FULL OUTER JOIN dept d ON s.did = d.dno AND d.dname <> 'CSE';

-- 4-19. CROSS JOIN between student and dept
SELECT s.sid, s.sname, d.dno, d.dname
FROM student s
CROSS JOIN dept d;
```

2. Execution and Output

Output shown is sample output of the programs above (Oracle Database 21c XE, SQL*Plus).

```
[4-1]
Table created.

 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DNO                                               NUMBER
 DNAME                                             VARCHAR2(20)

[4-2]
Table altered.

Table altered.

 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DNO                                       NOT NULL NUMBER
 DNAME                                     NOT NULL VARCHAR2(20)

[4-3]
Table created.

 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SID                                               NUMBER
 SNAME                                             VARCHAR2(30)
 DID                                               NUMBER

[4-4]
Table altered.

Table altered.

Table altered.

 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SID                                       NOT NULL NUMBER
 SNAME                                     NOT NULL VARCHAR2(30)
 DID                                               NUMBER

[4-5]
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
Commit complete.

DNO DNAME
--- -----
 10 CSE
 20 ME
 30 CE
 40 EEE
 50 ECE
 60 CSM
 70 CSD

7 rows selected.

[4-6]
10 rows created.

Commit complete.

SID SNAME DID
--- ----- ---
  1 Ravi   10
  2 Sita   10
  3 Arun   20
  4 Latha  30
  5 Kiran  40
  6 Meena  50
  7 Varun  20
  8 Divya  30
  9 Hari   10
 10 Anu
```

```
10 rows selected.
```

[4-7]

```
DID SID SNAME DNAME
--- --- ----- -----
 10   1 Ravi  CSE
 10   2 Sita  CSE
 20   3 Arun  ME
 30   4 Latha CE
 40   5 Kiran EEE
 50   6 Meena ECE
 20   7 Varun ME
 30   8 Divya CE
 10   9 Hari  CSE

9 rows selected.
```

[4-8]

```
SID SNAME DID DNO DNAME
--- ----- --- --- -----
  1 Ravi   10  10 CSE
  2 Sita   10  10 CSE
  3 Arun   20  20 ME
  4 Latha  30  30 CE
  5 Kiran  40  40 EEE
  6 Meena  50  50 ECE
  7 Varun  20  20 ME
  8 Divya  30  30 CE
  9 Hari   10  10 CSE

9 rows selected.
```

[4-9]

```
SID SNAME DID DNO DNAME
--- ----- --- --- -----
  3 Arun   20  20 ME
  4 Latha  30  30 CE
  5 Kiran  40  40 EEE
  6 Meena  50  50 ECE
  7 Varun  20  20 ME
  8 Divya  30  30 CE

6 rows selected.
```

[4-10]

```
DID SID SNAME DNAME
--- --- ----- -----
 10   1 Ravi  CSE
 10   2 Sita  CSE
 20   3 Arun  ME
 30   4 Latha CE
 40   5 Kiran EEE
 50   6 Meena ECE
 20   7 Varun ME
 30   8 Divya CE
 10   9 Hari  CSE
     10 Anu

10 rows selected.
```

[4-11]

```
DID SID SNAME DNAME
--- --- ----- -----
 10   1 Ravi  CSE
 10   2 Sita  CSE
 20   3 Arun  ME
 30   4 Latha CE
 40   5 Kiran EEE
 50   6 Meena ECE
 20   7 Varun ME
 30   8 Divya CE
 10   9 Hari  CSE
 60           CSM
 70           CSD

11 rows selected.
```

[4-12]

```
DID SID SNAME DNAME
--- --- ----- -----
 10   1 Ravi  CSE
 10   2 Sita  CSE
 20   3 Arun  ME
 30   4 Latha CE
 40   5 Kiran EEE
 50   6 Meena ECE
 20   7 Varun ME
 30   8 Divya CE
 10   9 Hari  CSE
     10 Anu
 60           CSM
 70           CSD

12 rows selected.
```

[4-13]

```
SID SNAME DID DNO DNAME
--- ----- --- --- -----
  1 Ravi   10  10 CSE
  2 Sita   10  10 CSE
  3 Arun   20  20 ME
  4 Latha  30  30 CE
  5 Kiran  40  40 EEE
  6 Meena  50  50 ECE
  7 Varun  20  20 ME
  8 Divya  30  30 CE
  9 Hari   10  10 CSE
 10 Anu

10 rows selected.
```

[4-14]

```
SID SNAME DID DNO DNAME
--- ----- --- --- -----
  1 Ravi   10  10 CSE
  2 Sita   10  10 CSE
  3 Arun   20  20 ME
  4 Latha  30  30 CE
  5 Kiran  40  40 EEE
  6 Meena  50  50 ECE
  7 Varun  20  20 ME
  8 Divya  30  30 CE
  9 Hari   10  10 CSE
               60 CSM
               70 CSD

11 rows selected.
```

[4-15]

```
SID SNAME DID DNO DNAME
--- ----- --- --- -----
  1 Ravi   10  10 CSE
  2 Sita   10  10 CSE
  3 Arun   20  20 ME
  4 Latha  30  30 CE
  5 Kiran  40  40 EEE
  6 Meena  50  50 ECE
  7 Varun  20  20 ME
  8 Divya  30  30 CE
  9 Hari   10  10 CSE
 10 Anu
               60 CSM
               70 CSD

12 rows selected.
```

[4-16]

```
SID SNAME DID DNO DNAME
--- ----- --- --- -----
  1 Ravi   10
  2 Sita   10
  3 Arun   20  20 ME
  4 Latha  30  30 CE
  5 Kiran  40  40 EEE
  6 Meena  50  50 ECE
  7 Varun  20  20 ME
  8 Divya  30  30 CE
  9 Hari   10
```

```
 10 Anu

10 rows selected.
```

[4-17]

```
SID SNAME DID DNO DNAME
--- ----- --- --- -----
  3 Arun   20  20 ME
  4 Latha  30  30 CE
  5 Kiran  40  40 EEE
  6 Meena  50  50 ECE
  7 Varun  20  20 ME
  8 Divya  30  30 CE
               10 CSE
               60 CSM
               70 CSD

9 rows selected.
```

[4-18]

```
SID SNAME DID DNO DNAME
--- ----- --- --- -----
  1 Ravi   10
  2 Sita   10
  3 Arun   20  20 ME
  4 Latha  30  30 CE
  5 Kiran  40  40 EEE
  6 Meena  50  50 ECE
  7 Varun  20  20 ME
  8 Divya  30  30 CE
  9 Hari   10
 10 Anu
               10 CSE
               60 CSM
               70 CSD

13 rows selected.
```

[4-19]

```
SID SNAME DNO DNAME
--- ----- --- -----
  1 Ravi   10 CSE
  1 Ravi   20 ME
  1 Ravi   30 CE
  1 Ravi   40 EEE
  1 Ravi   50 ECE
  1 Ravi   60 CSM
  1 Ravi   70 CSD
  2 Sita   10 CSE
  2 Sita   20 ME
  2 Sita   30 CE
  2 Sita   40 EEE
  2 Sita   50 ECE
  2 Sita   60 CSM
  2 Sita   70 CSD
  3 Arun   10 CSE
  3 Arun   20 ME
  3 Arun   30 CE
  3 Arun   40 EEE
  3 Arun   50 ECE
  3 Arun   60 CSM
  3 Arun   70 CSD
  4 Latha  10 CSE
  4 Latha  20 ME
  4 Latha  30 CE
  4 Latha  40 EEE
  4 Latha  50 ECE
  4 Latha  60 CSM
  4 Latha  70 CSD
  5 Kiran  10 CSE
  5 Kiran  20 ME
  5 Kiran  30 CE
  5 Kiran  40 EEE
  5 Kiran  50 ECE
  5 Kiran  60 CSM
  5 Kiran  70 CSD
  6 Meena  10 CSE
  6 Meena  20 ME
  6 Meena  30 CE
  6 Meena  40 EEE
  6 Meena  50 ECE
```

```
  6 Meena  60 CSM
  6 Meena  70 CSD
  7 Varun  10 CSE
  7 Varun  20 ME
  7 Varun  30 CE
  7 Varun  40 EEE
  7 Varun  50 ECE
  7 Varun  60 CSM
  7 Varun  70 CSD
  8 Divya  10 CSE
  8 Divya  20 ME
  8 Divya  30 CE
  8 Divya  40 EEE
  8 Divya  50 ECE
  8 Divya  60 CSM
  8 Divya  70 CSD
  9 Hari   10 CSE
  9 Hari   20 ME
  9 Hari   30 CE
  9 Hari   40 EEE
  9 Hari   50 ECE
  9 Hari   60 CSM
  9 Hari   70 CSD
 10 Anu    10 CSE
 10 Anu    20 ME
 10 Anu    30 CE
 10 Anu    40 EEE
 10 Anu    50 ECE
 10 Anu    60 CSM
 10 Anu    70 CSD

70 rows selected.
```